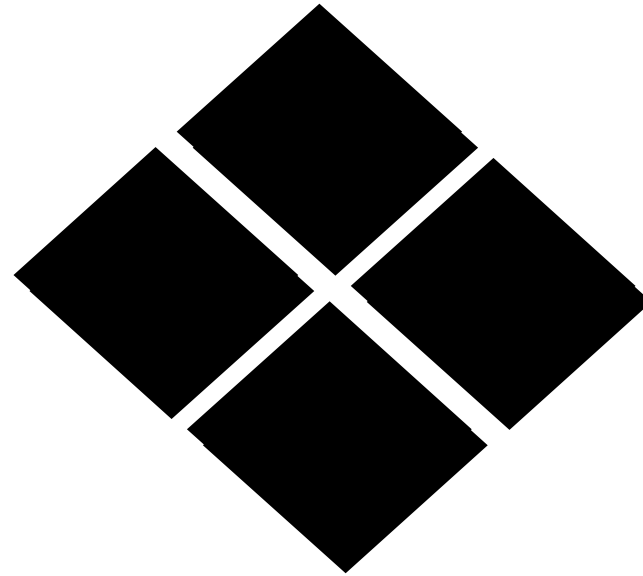
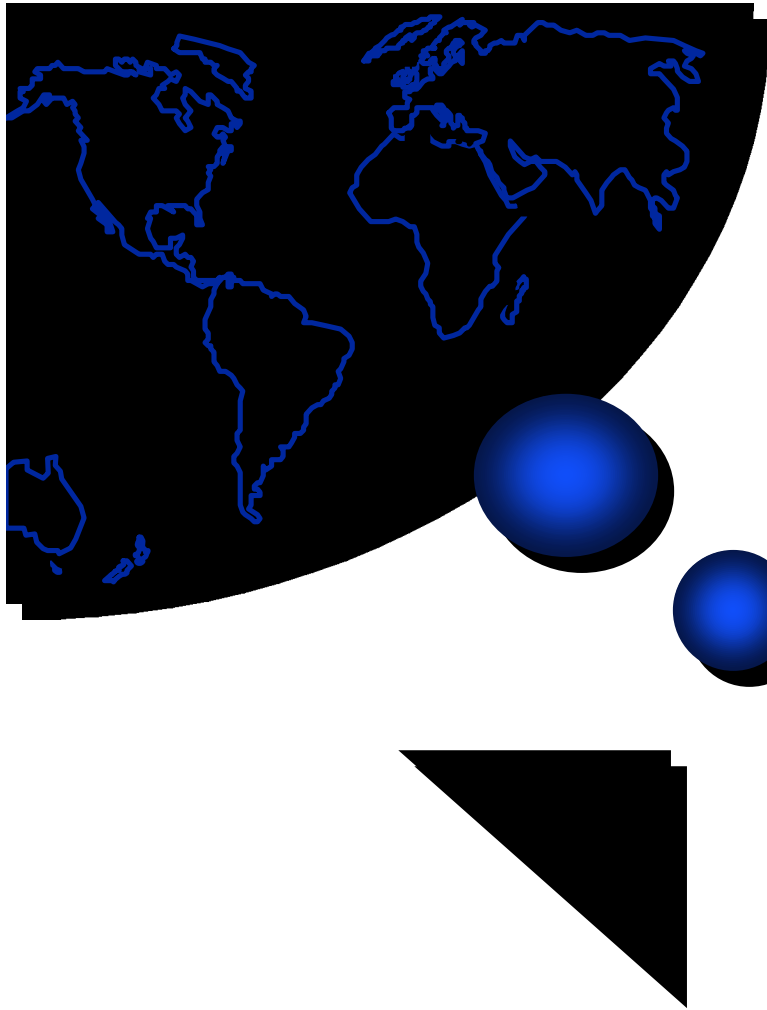




An Overview of IEEE Software Engineering Standards and Knowledge Products

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Objectives



-
- Provide an introduction to The IEEE Software Engineering Standards Committee (SESC)
 - Provide an overview of the current state and future direction of IEEE Software Engineering Standards and knowledge products
 - ◆ IEEE Software Engineering Standards Collection
 - ◆ Software Engineering Competency Recognition Program
 - ◆ Standards-Based Training
 - Discuss how you can participate in software engineering standardization efforts



The IEEE Software Engineering Standards Committee (SESC)

<http://computer.org/standard/sesc/>



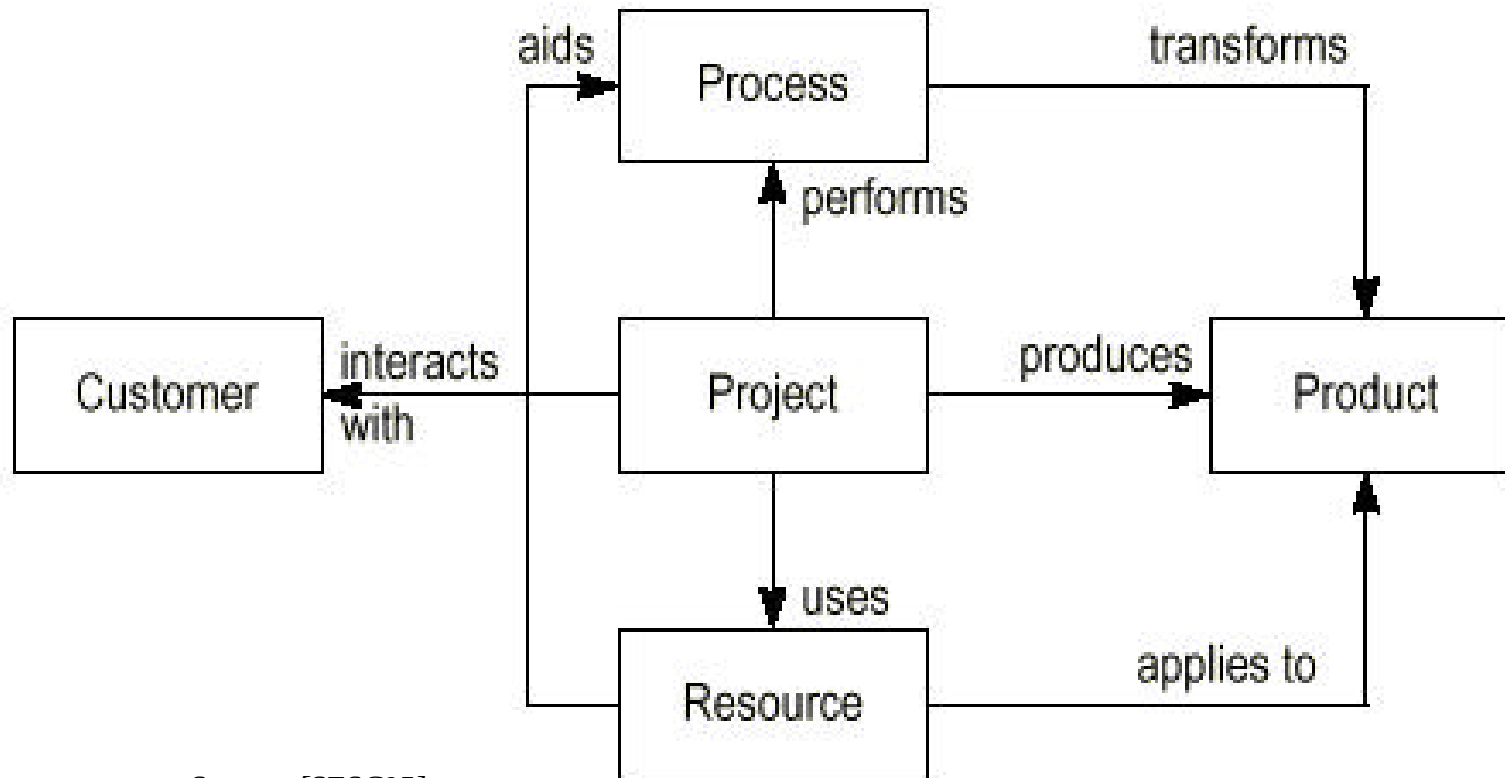
The SESC Vision



-
- The leading supplier and promoter of a family of software engineering standards and related products and services.



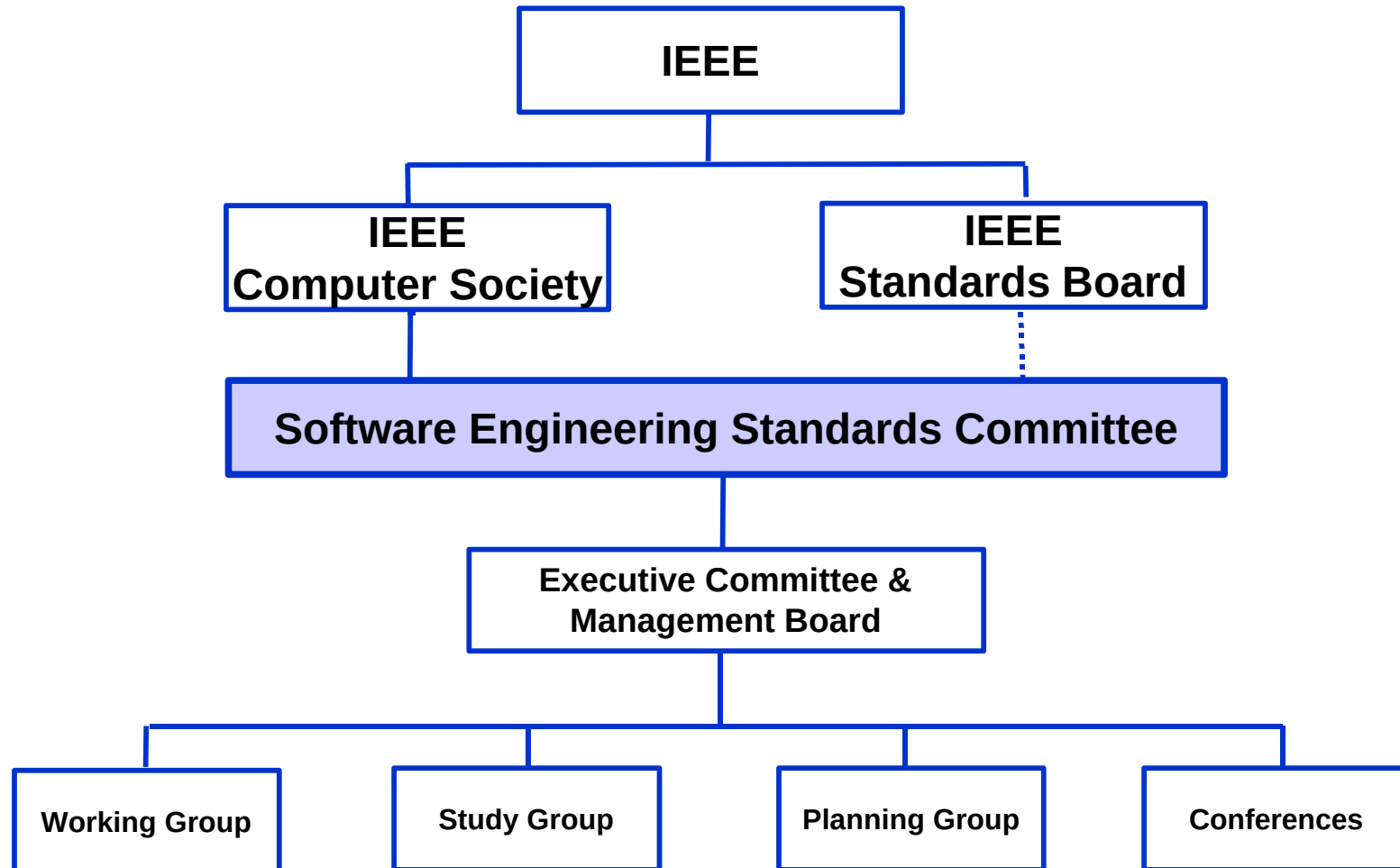
Software Engineering: An Object View



Source: [SESC95]

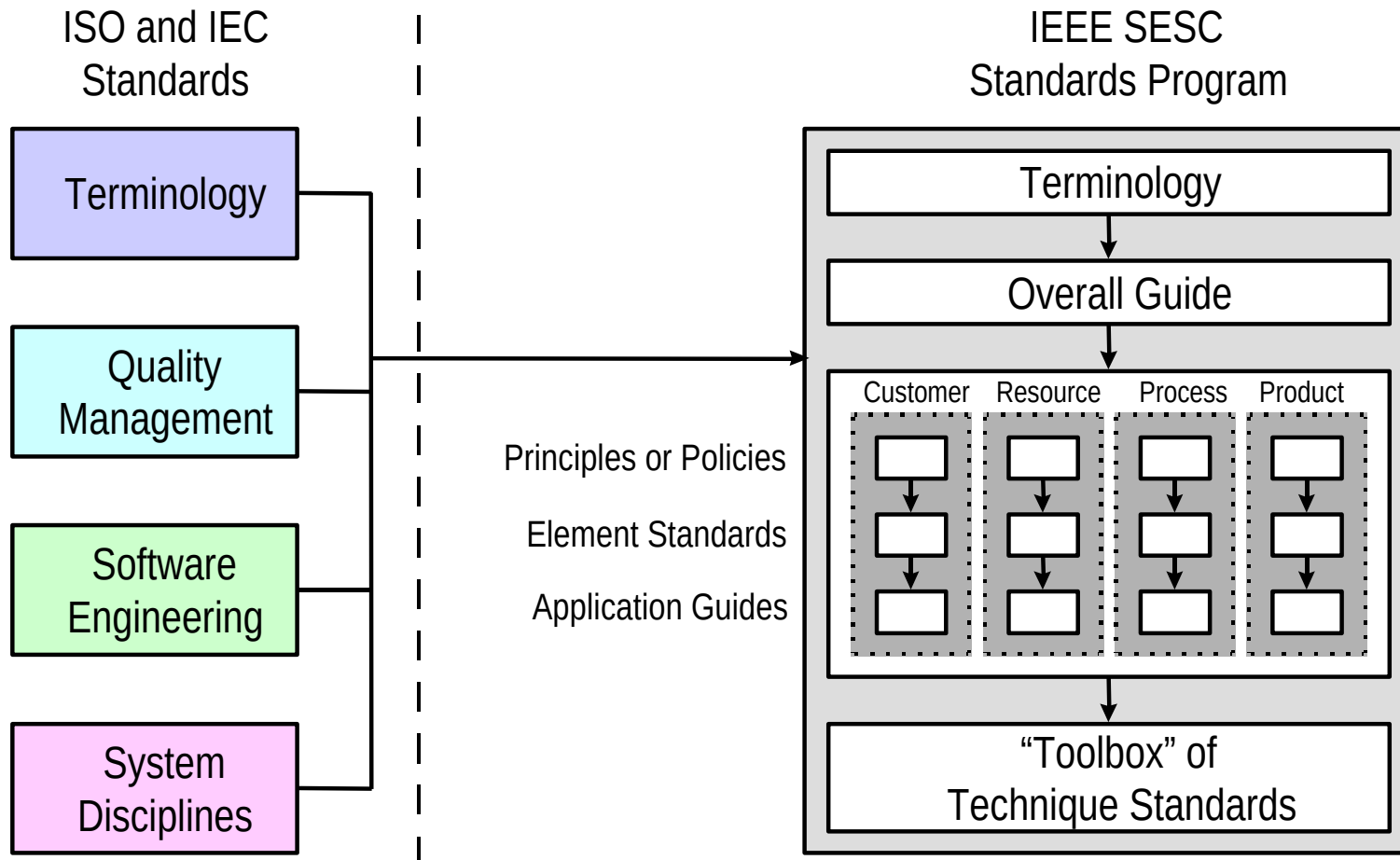


SESC in the IEEE Structure





SESC Strategic Program Model



Source: [SESC95]



The IEEE Software Engineering Standards Collection

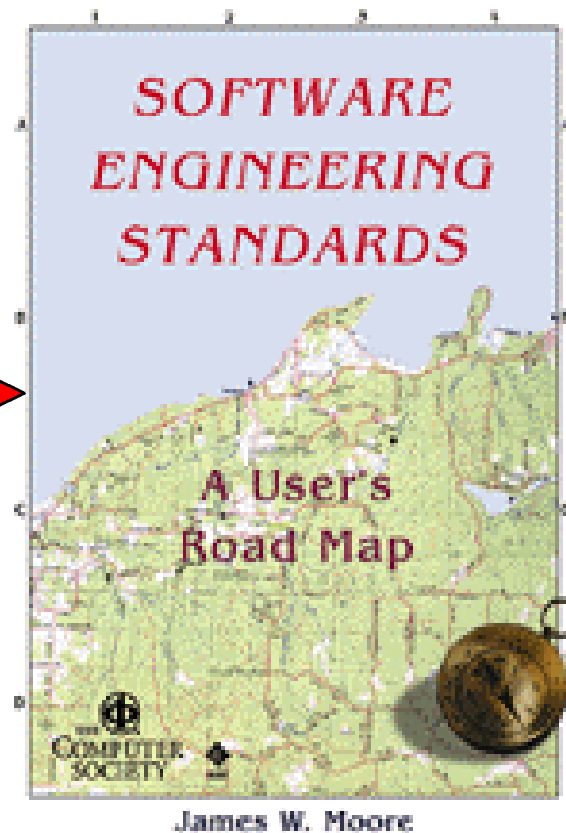
<http://standards.ieee.org/catalog/softwareset.html>



The 2000 Software Engineering Standards Collection



- Forty-six Standards
 - ◆ Customer & Terminology
 - ◆ Process
 - ◆ Product
 - ◆ Resource & Technique
- Overall guide
 - ◆ Several “views”
 - Context
 - Object
 - Normative intent
 - Provider and subject
 - ◆ Relationships among standards



James W. Moore

Source: [Moore97]



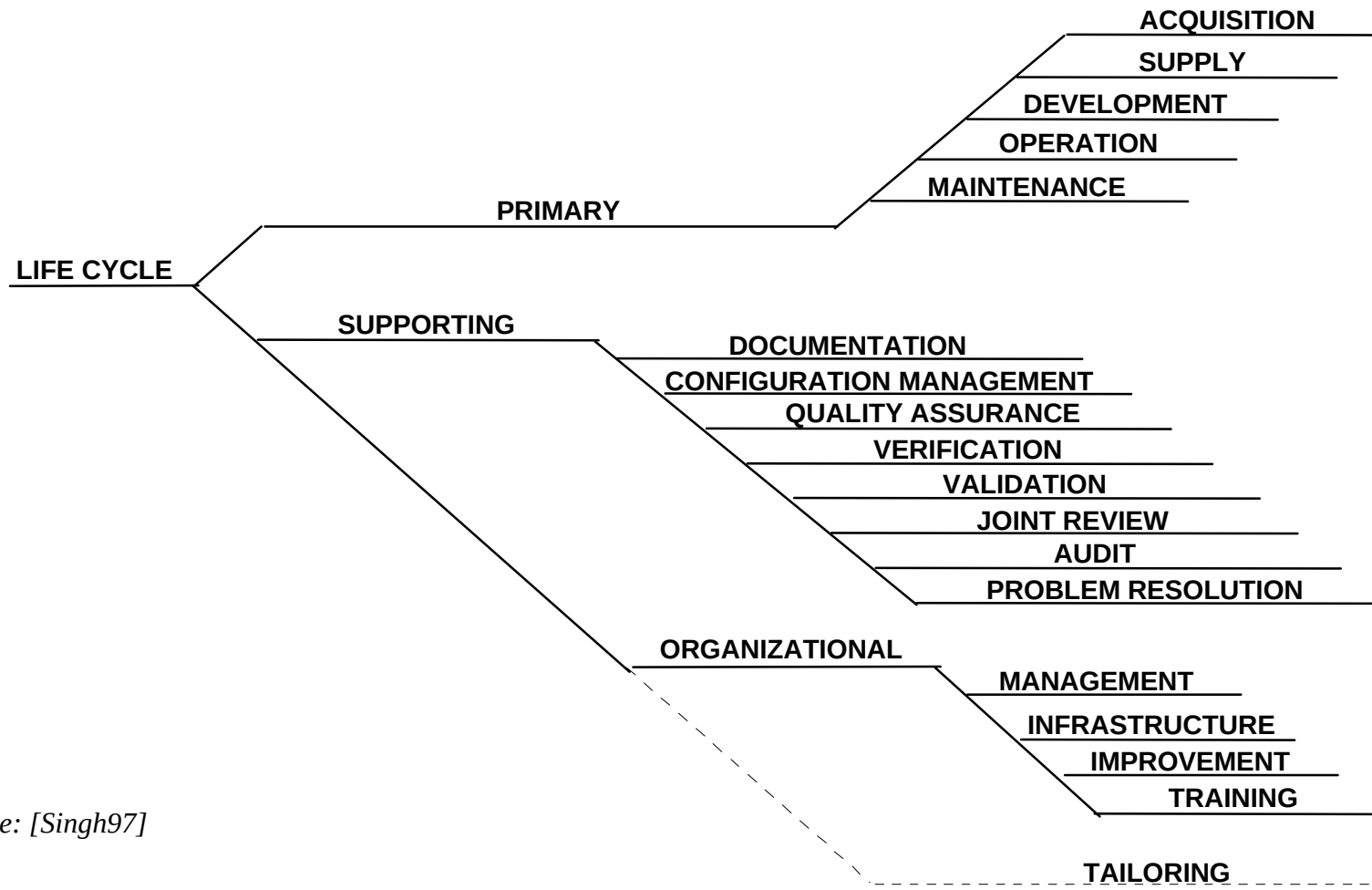
IEEE/EIA 12207: The Life Cycle Process Framework



- IEEE/EIA 12207, Standard for Information Technology – Software Life Cycle Processes
 - ◆ Addresses the complete software engineering life cycle, from acquisition and supply, through development, to operations and maintenance
 - ◆ Provides a process framework upon which an organization can build its enterprise-level life cycle processes
 - ◆ These enterprise-level processes are then tailored into projects, in order to meet specific project-level requirements.



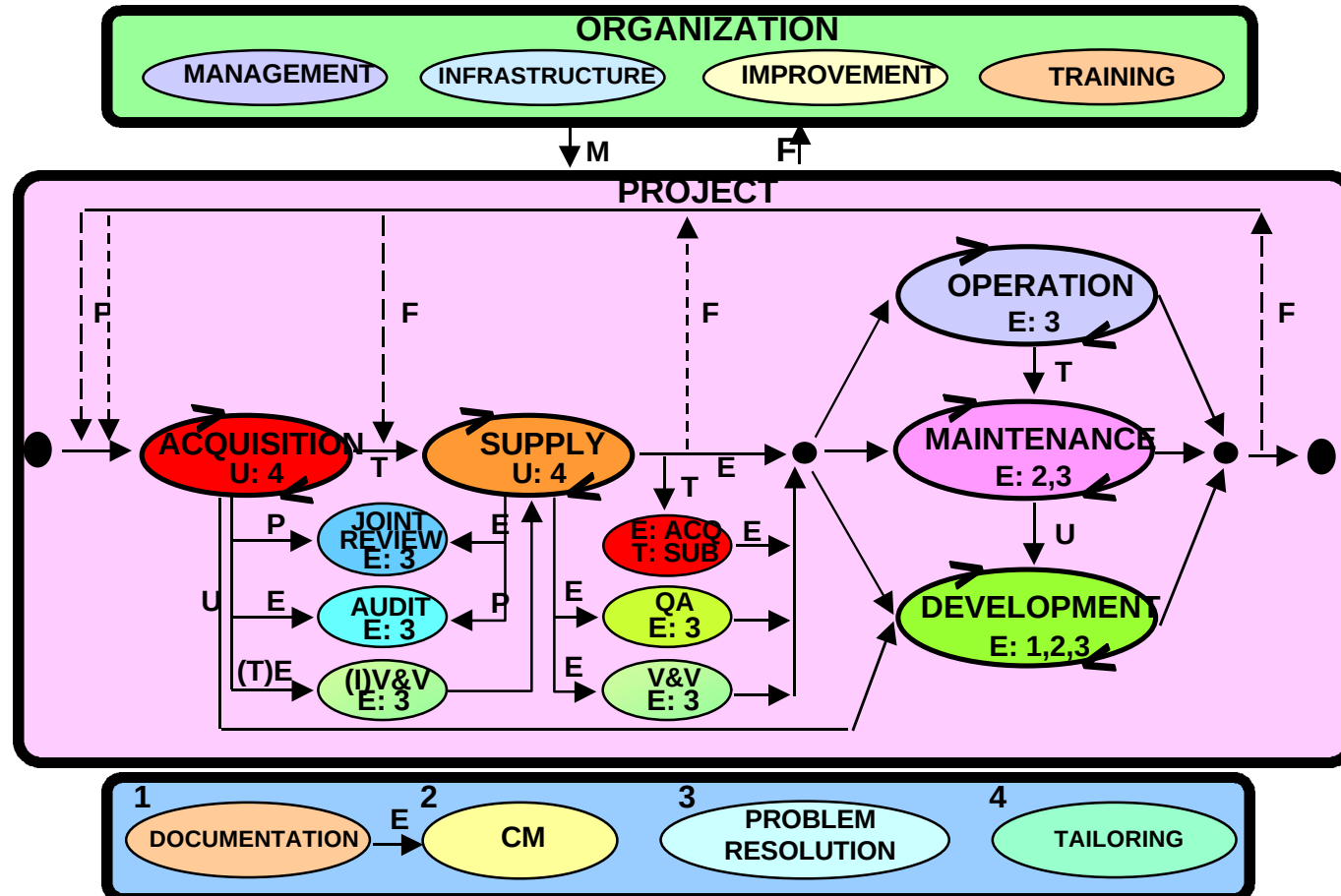
IEEE/EIA 12207 Process Tree



Source: [Singh97]



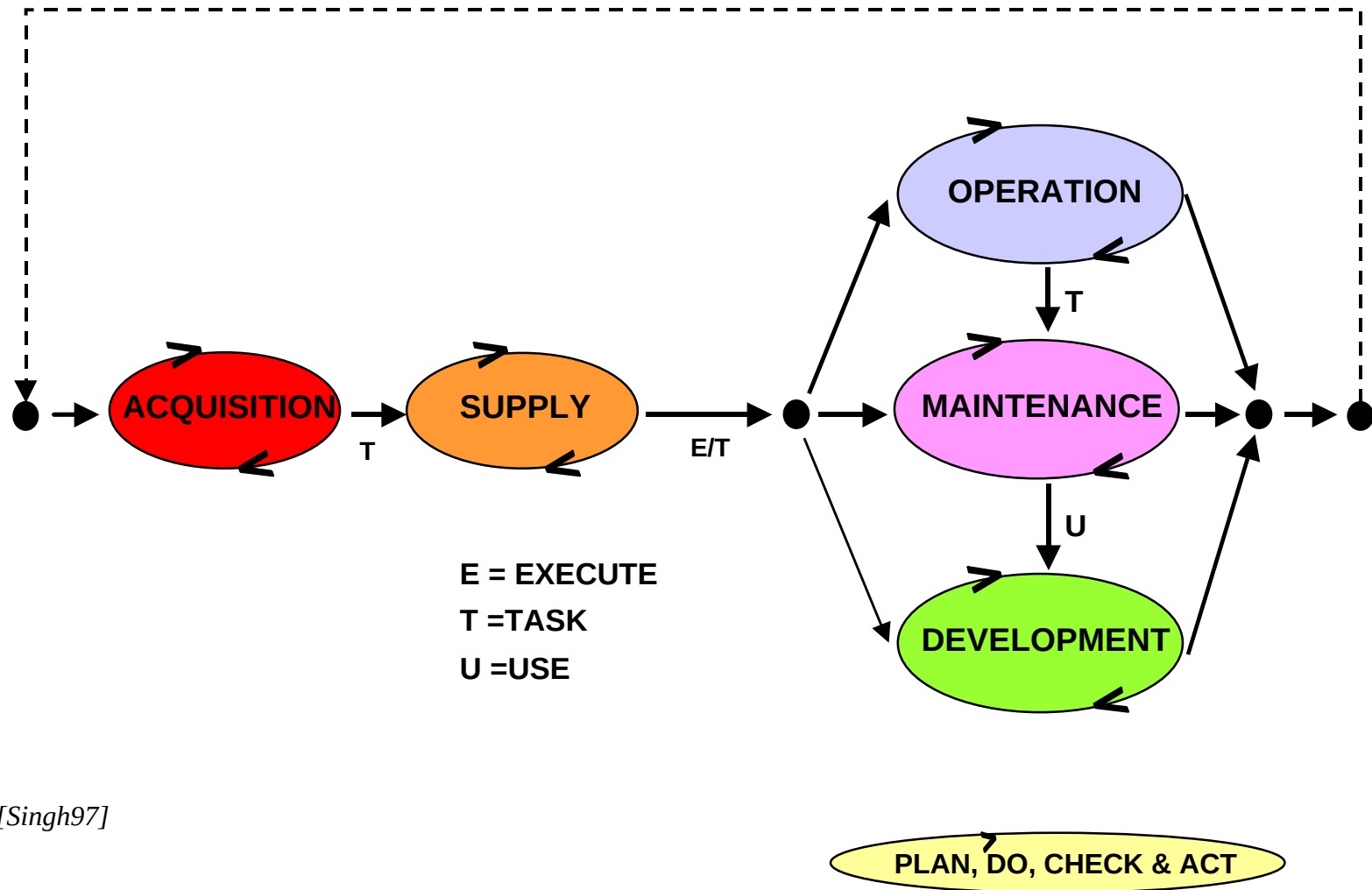
12207 Process Flow



Source: [Singh97]



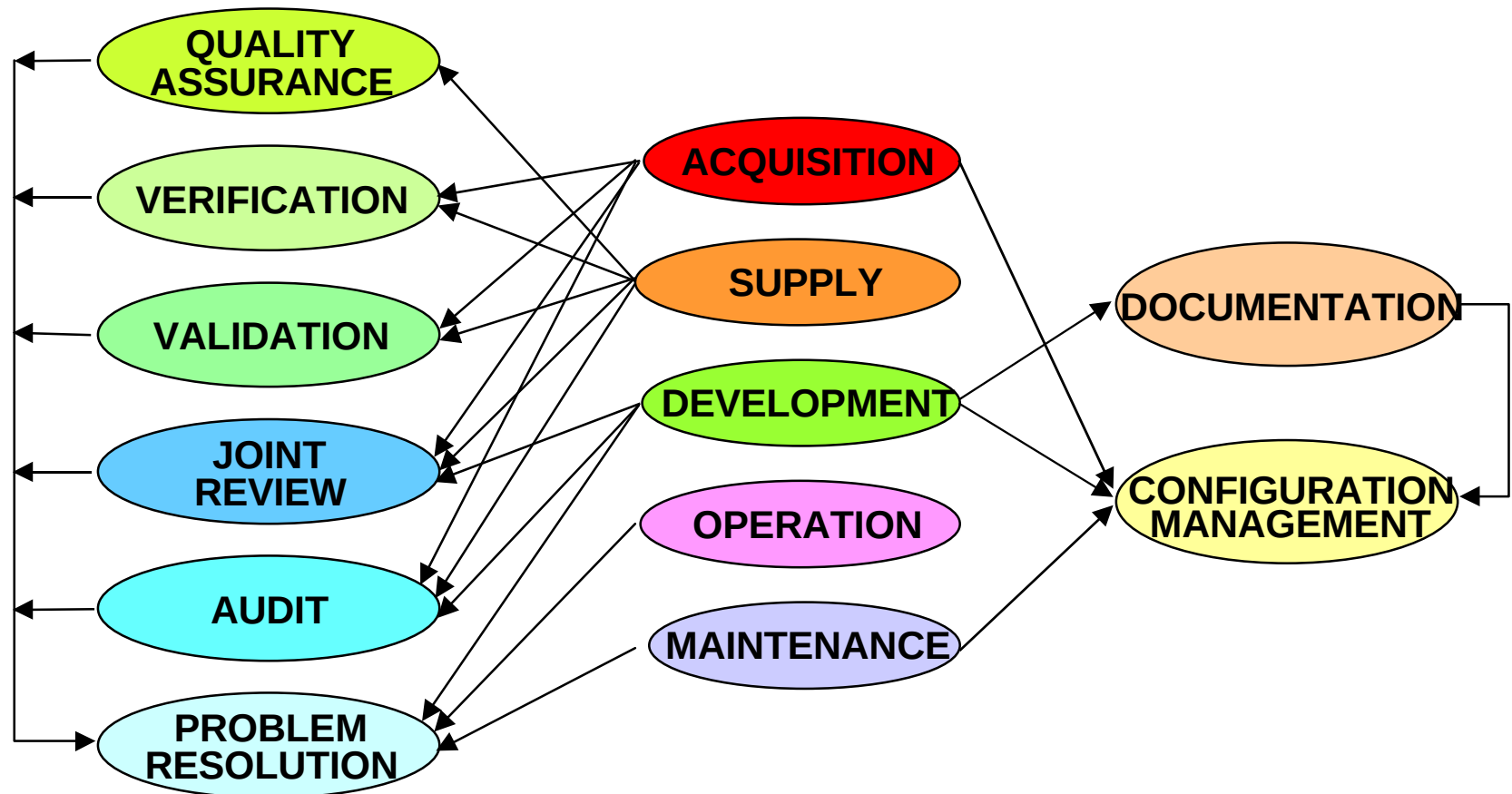
Primary Process Flow



Source: [Singh97]



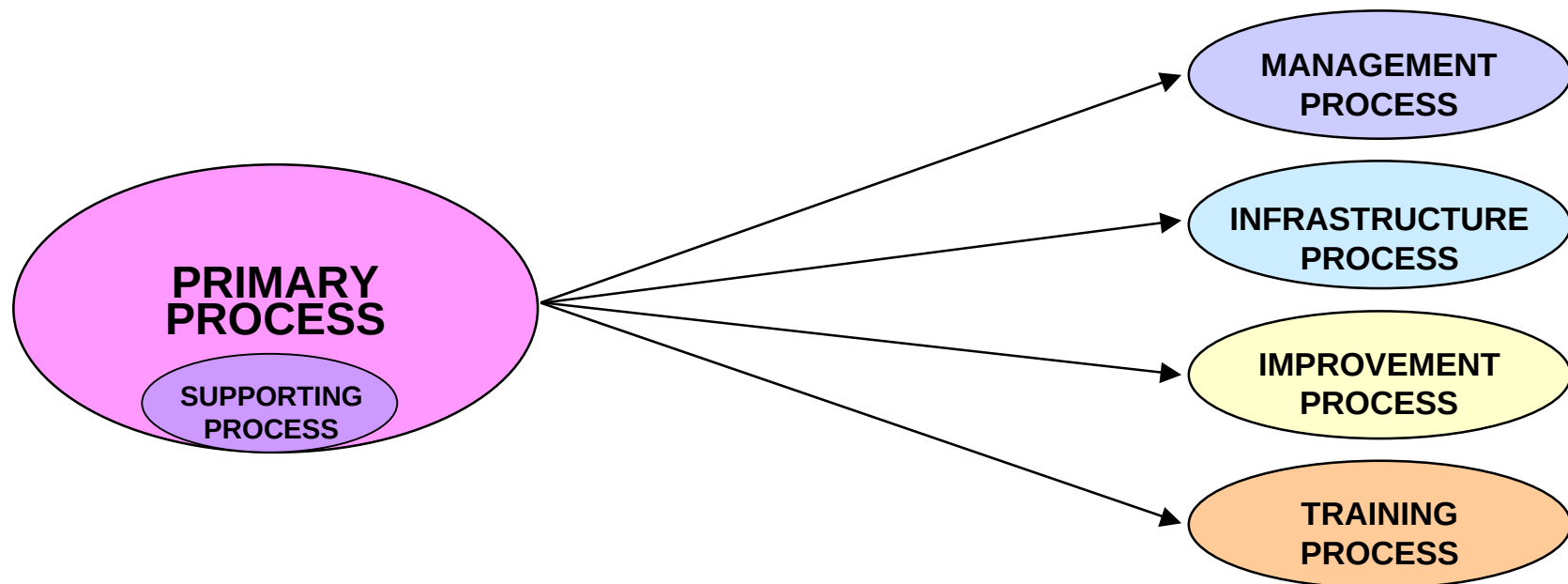
Supporting Process Flow



Source: [Singh97]



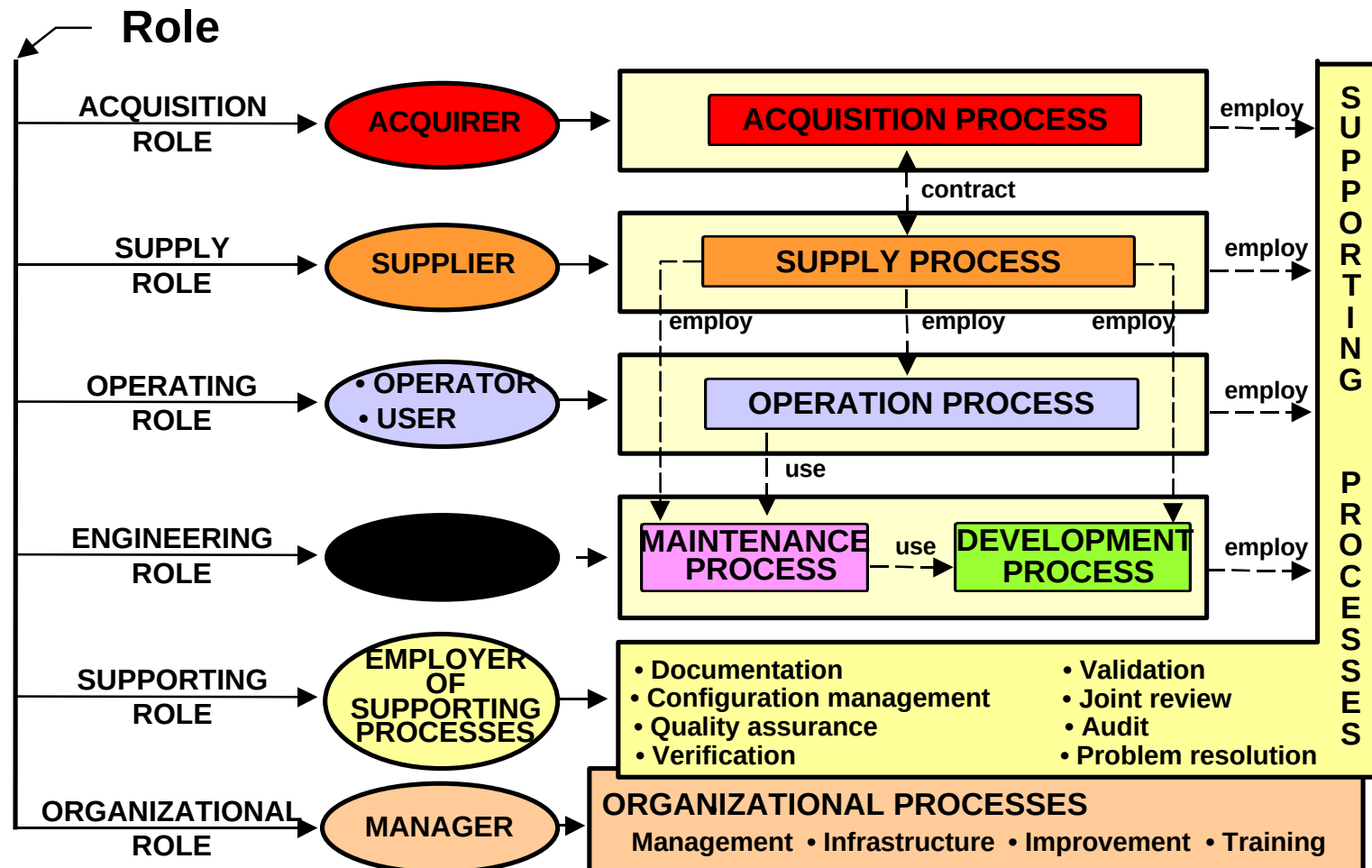
Organizational Process Flow



Source: [Singh97]



12207 Process Roles



Source: [Singh97]



Role Definitions



- Acquirer:
 - ◆ an organization that acquires or procures a system, software product or software service from a supplier
- Supplier:
 - ◆ an organization that enters into a contract with the acquirer for the supply of a system, software product or software service under the terms of the contract
- Operator:
 - ◆ an organization that operates the system



Role Definitions - 2



- Developer:
 - ◆ an organization that performs development activities (including requirements analysis, design, testing through acceptance) during the software life cycle process
- Maintainer:
 - ◆ an organization that performs maintenance activities
- Supporting Process Performer and Manager are undefined



IEEE/EIA 12207 Document Structure



- IEEE/EIA 12207.0-1996, Software Life Cycle Processes
 - ◆ Industry adoption of ISO/IEC 12207-1995
- IEEE/EIA 12207.1-1997, Life Cycle Data
 - ◆ Industry guide to life cycle data
- IEEE/EIA 12207.2-1997, Implementation Considerations
 - ◆ Industry guide to implementation of the life cycle processes contained in 12207.0



IEEE/EIA 12207.0

Document Structure



- Foreword to IEEE/EIA 12207.0-1996
- ISO/IEC 12207-1995
 - ◆ Introduction
 - ◆ Foreword
 - ◆ Clause 1 - Scope
 - ◆ Clause 2 - Normative references
 - ◆ Clause 3 - Definitions
 - ◆ Clause 4 - Application of this International Standard
 - ◆ Clause 5 - Primary life cycle processes
 - ◆ Clause 6 - Supporting processes
 - ◆ Clause 7 - Organizational life cycle processes



IEEE/EIA 12207.0 Annexes



- ISO/IEC 12207-1995 Annexes
 - ◆ A - Tailoring process
 - ◆ B - Guidance on tailoring
 - ◆ C - Guidance on processes and organizations
 - ◆ D - Bibliography
- Additional IEEE/EIA 12207.0 Annexes
 - ◆ E - Basic concepts of ISO/IEC 12207
 - ◆ F - Compliance
 - ◆ G - Life cycle processes objectives
 - ◆ H - Life cycle data objectives
 - ◆ I - Relationships
 - ◆ J - Errata



IEEE/EIA 12207.1

Document Structure



- ◆ Introduction
- ◆ Clause 1 - Scope
- ◆ Clause 2 - Normative references
- ◆ Clause 3 - Definitions
- ◆ Clause 4 - Life cycle data
 - Clause 4.1 Overview
 - Clause 4.2 Life cycle data objectives
 - Clause 4.3 Information item matrix
 - Clause 4.4 Compliance
- ◆ Clause 5 - Generic information item content guidelines
- ◆ Clause 6 - Specific information item content guidelines
- ◆ Annex A - References



IEEE/EIA 12207.2

Document Structure



- Foreword
- Introduction
- Clause 1 - Scope
- Clause 2 - Normative references
- Clause 3 - Definitions
- Clause 4 - Application
- Clause 5 - Primary life cycle processes
- Clause 6 - Supporting processes
- Clause 7 - Organizational life cycle processes



IEEE/EIA 12207.2 Annexes



- IEEE/EIA 12207 Annexes
 - ◆ A - IEEE/EIA 12207.0 Annex A - Tailoring process
 - ◆ B - IEEE/EIA 12207.0 Annex F - Compliance
 - ◆ C - IEEE/EIA 12207.0 Annex G - Life cycle processes objectives
 - ◆ D - IEEE/EIA 12207.0 Annex H - Life cycle data objectives
 - ◆ E - IEEE/EIA 12207.0 Annex J - Errata



IEEE/EIA 12207.2 Annexes - 2



- Additional IEEE/EIA 12207.2 Annexes
 - ◆ F - Use of reusable software products
 - ◆ G - Candidate joint management reviews
 - ◆ H - Software measurement categories
 - ◆ I - Guidance on development strategies and build planning
 - ◆ J - Category and priority classifications for problem reporting
 - ◆ K - Software product evaluations
 - ◆ L - Risk management
 - ◆ M - Life cycle processes references



Supporting Standards for High Integrity Software



- IEEE/EIA 12207 relies upon other standards to fill in the details regarding the activities supporting life cycle processes.
- In the case of high integrity software, several additional software engineering standards are of interest.



Customer and Terminology



- 610.12, Standard Glossary of Software Engineering Terminology
- **1062, Recommended Practice for Software Acquisition**
- **1220, Standard for Application and Management of the Systems Engineering Process**
- **1228, Standard for Software Safety Plans**
- **1233, Guide for Developing System Requirements Specifications**
- **1362, Guide for Concept of Operations Document**
- **12207, Software Life Cycle Processes**
- **12207.1, Guide to Software Life Cycle Processes—Life Cycle Data**
- **12207.2, Guide to Software Life Cycle Processes—Implementation Considerations**

■ = High Integrity Systems Related



Process



-
- 730, Standard for Software Quality Assurance Plans
 - 730.1, Guide for Software Quality Assurance Planning
 - 828, Standard for Software Configuration Management Plans
 - **1008, Standard for Software Unit Testing**
 - **1012, Standard for Software Verification and Validation**
 - **1012a, Software Verification and Validation Content Map to IEEE/EIA 12207.1**
 - 1028, Standard for Software Reviews
 - 1042, Guide to Software Configuration Management
 - 1045, Standard for Software Productivity Metrics
 - 1058, Standard for Software Project Management Plans
 - 1059, Guide for Software Verification and Validation Plans
 - 1074, Standard for Developing Software Life Cycle Processes
 - 1219, Standard for Software Maintenance
 - 1490, A Guide to the Program Management Body of Knowledge
- = High Integrity Systems Related



Process - 2



- J-STD-016-1995, (EIA/IEEE) Interim Standard for Information Technology - Software Life Cycle Processes - Software Development - Acquirer-Supplier Agreement
- 1517-1999, Standard for Information Technology - Software Life Cycle Processes - Reuse Processes
- **P1540, D7.0, Draft Standard for Software Life Cycle Processes - Risk Management**

■ = High Integrity Systems Related



Product



- **982.1, Standard Dictionary of Measures to Produce Reliable Software**
- **982.2, Guide for the Use of Standard Dictionary of Measures to Produce Reliable Software**
- 1061, Standard for a Software Quality Metrics Methodology
- 1063, Standard for Software User Documentation
- 1465, IEEE Standard Adoption of ISO/IEC 12119: 1994 (E) International Standard--Information Technology - Software Packages - Quality Requirements and Testing
- 14143.1, Approved Draft - Standard Adoption of ISO/IEC 1443-1:1998 - Information Technology - Software Measurement - Functional Size Measurement - Part 1: Definition of Concepts

■ = High Integrity Systems Related



Resource and Technique



- **829, Standard for Software Test Documentation**
- **830, Recommended Practice for Software Requirements Specifications**
- **1016, Recommended Practice for Software Design Descriptions**
- 1044, Standard Classification for Software Anomalies
- 1044.1, Guide to Classification for Software Anomalies
- 1320.1, Syntax and Semantics for IDEF0
- 1320.2, Syntax and Semantics for IDEF1X97 (IDEFObject)
- 1348, Recommended Practice for the Adoption of CASE Tool
- 1420.1, Software Reuse—Data Model for Reuse Library Interoperability: Basic Interoperability Data Model
- 1420.1a, Software Reuse—Data Model for Reuse Library Interoperability: Asset Certification Framework
- 1420.1b-1999, Trial Use Supplement - Software Reuse—Data Model for Reuse Library Interoperability: Data Model for Reuse Library Interoperability: Intellectual Property Rights Framework

■ = High Integrity Systems Related
ASQ Section 509 SSIG Meeting, 8 November 2000



Resource and Technique - 2

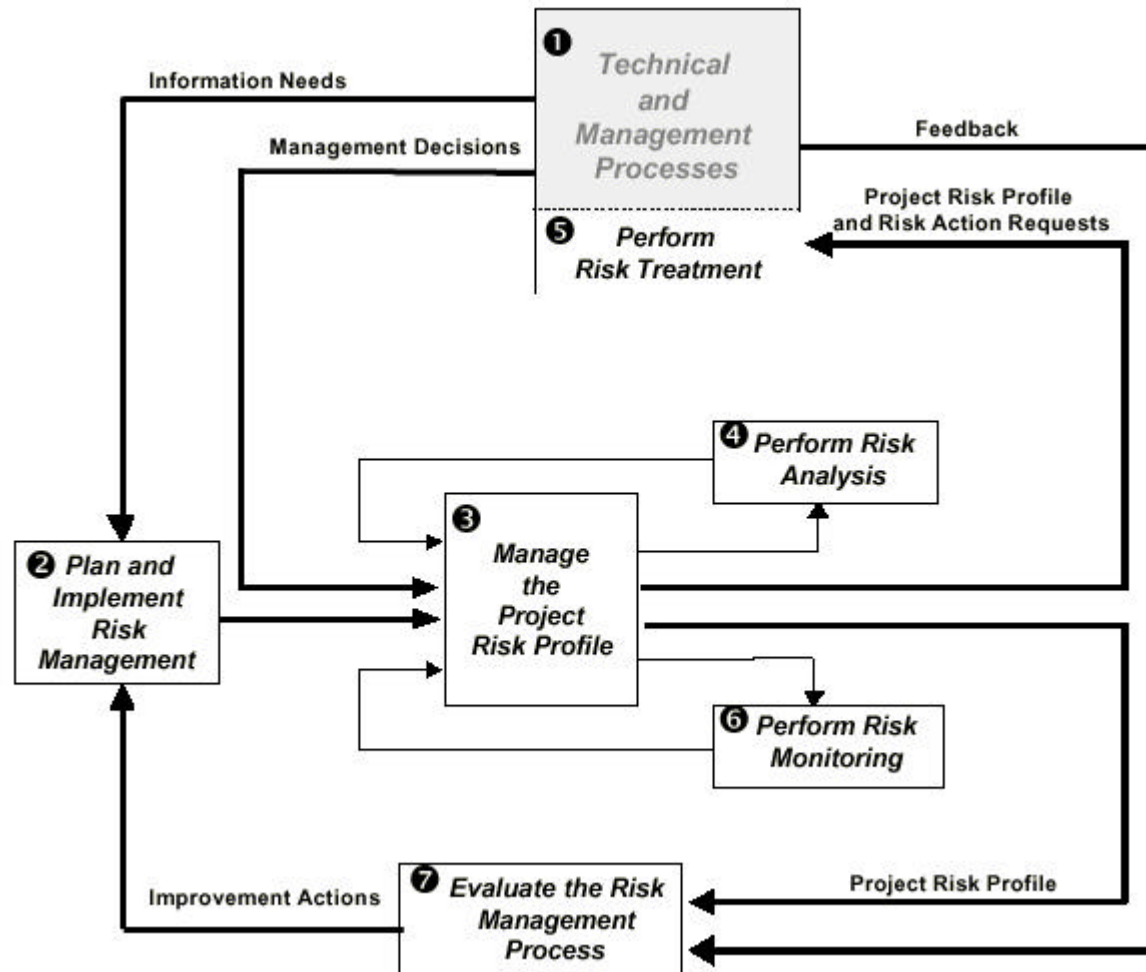


- 1430, Guide for Software Reuse - Concept of Operations for Interoperating Reuse Libraries
- 1462, Guide for the Evaluation and Selection of CASE Tools
- **P1471, Recommended Practice For Architectural Description of Software Intensive Systems**

■ = High Integrity Systems Related

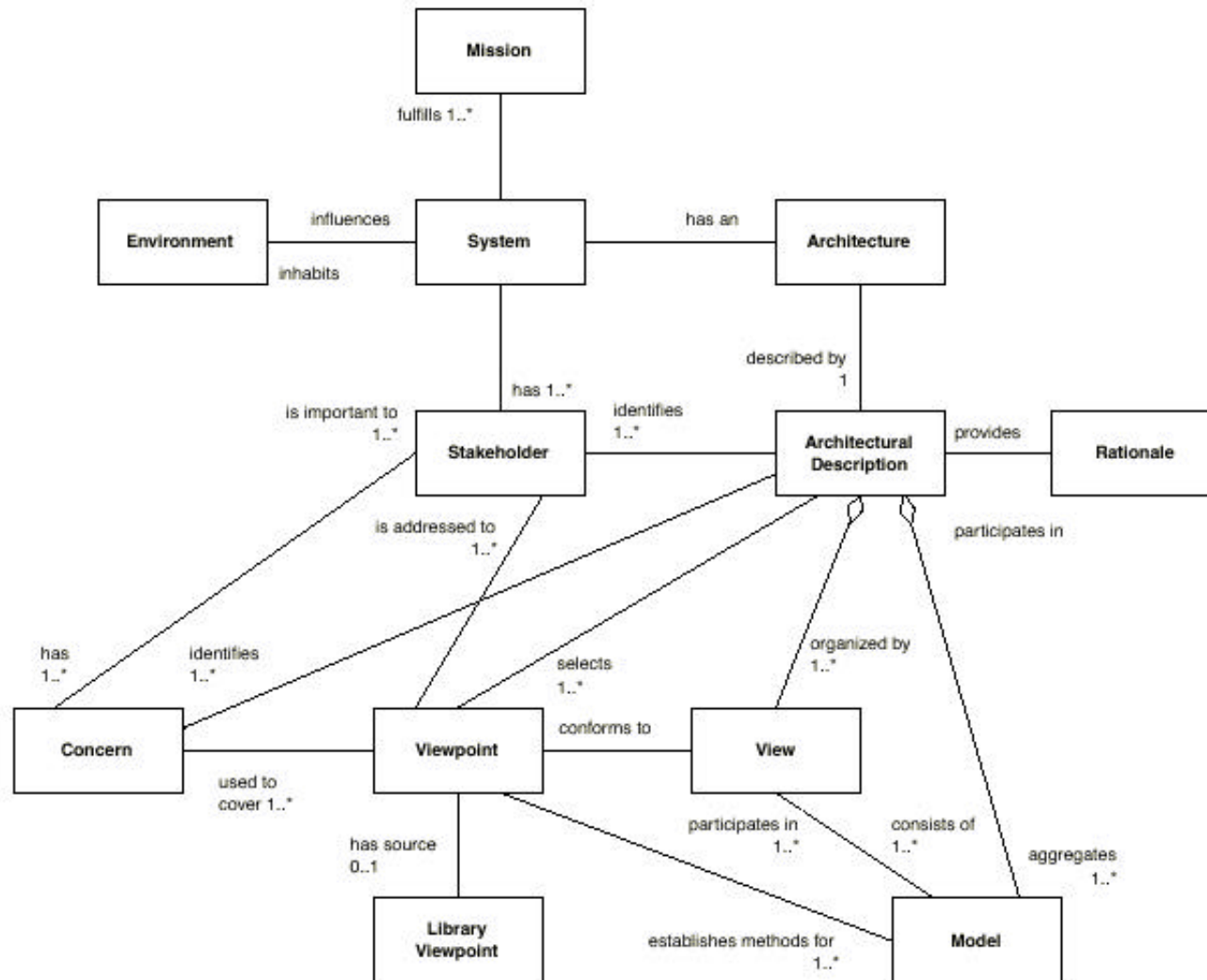


IEEE 1540: Software Risk Management - Process Model





IEEE 1471: Recommended Practice for Architectural Description of Software-Intensive Systems - Conceptual Model





Software Engineering Competency: Professionalizing Software Engineering



The Three Components of Engineering Competency



- A defined Body of Knowledge
- A Code of Practice
- Competency recognition



Guide to the Software Engineering Body of Knowledge



- Objectives
 - ◆ Better characterize the discipline of Software Engineering
 - ◆ Provide a consistent view Software Engineering as an engineering discipline

<http://www.swebok.org>



IEEE Software Engineering Competency Recognition Program



- Goals
 - ◆ Identify qualified professionals
 - ◆ Ensure recognition of expertise
 - ◆ Assist in professional development
 - ◆ Establish professional practice standards
 - ◆ Protect public
 - ◆ Enable professionals to stay current

Source: [IEEE99]



IEEE Software Engineering Competency Recognition Program - 2



- Roles
 - ◆ software engineering practitioner
 - ◆ software project manager
 - ◆ software systems architect
 - ◆ supporter (e.g. CM, QA, etc.)



Standards-Based Training



- Skills training in the “Code of Practice”
 - ◆ tailorable course outlines
 - ◆ completion certificates
- Pilot training program
 - ◆ State of California
 - ◆ New York City Transit Authority
 - ◆ Delta Airlines
- Twenty-three courses were delivered last year to 500 attendees
- Five universities contracted to teach courses



IEEE Software Engineering Standards Committee

Our Future and Yours



SESC objectives for the New Millennium



- A consistent collection of Software Engineering Standards to support process definition and product development, that improve the quality of delivered software and software-intensive systems
- Development and delivery of Standards-based training to improve skills
- Feedback mechanisms to capture experience in standards usage
- A conformance program for the organizational implementation of SESC standards

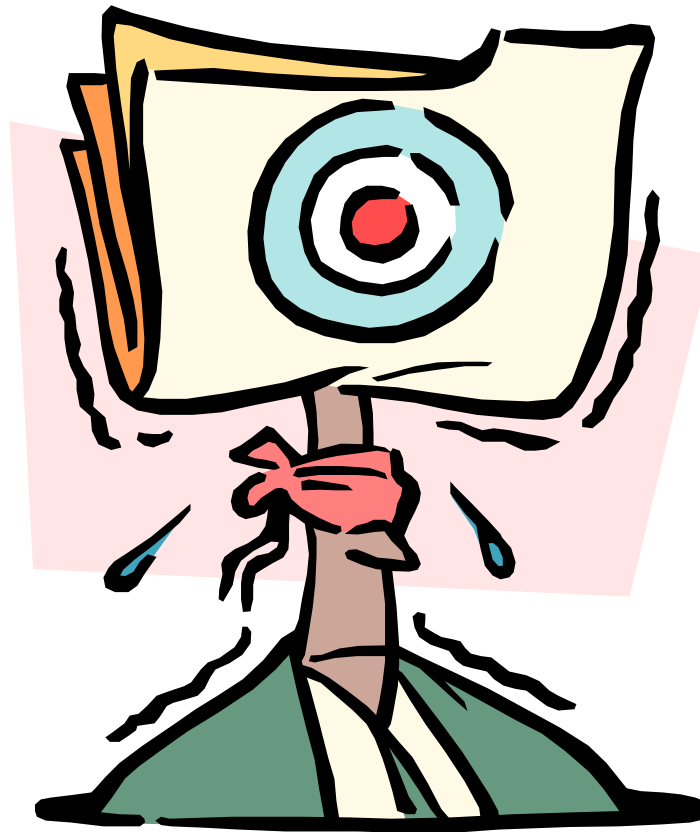


How You Can Participate



-
- Join the IEEE Computer Society
(at <http://www.computer.org>)
 - Join the IEEE Software Engineering Standards Committee (at <http://www.tcse.org>)
 - ◆ Lead or participate in Working Groups developing or revising Standards
 - ◆ Lead or participate in Study Groups investigating new areas for standardization
 - ◆ Participate in SESC special projects
 - ◆ Become part of the SESC balloting pool (IEEE Standards Association membership required)

Questions





For more information . . .



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References



-
- [IEEE99] IEEE Computer Society, “Business Plan for the Software Engineering Competency Recognition Program”, May1999
 - [Moore97] James W. Moore, *Software Engineering Standards: A User's Road Map*, IEEE Computer Society Press, Los Alamitos, CA, 1997.
 - [SESC95] SESC Business Planning Group, “Vision 2000 Strategy Statement (Final Draft),” v0.9, SESC/BPG-002, August 20, 1995.
 - [Singh97] Raghu Singh, *An Introduction to International Standards ISO/IEC 12207, Software Life Cycle Processes*, 1997.